17	**D**	Since intensity, I varies proportionately with (amplitude, A)^2 Let each unit for displacement be y. $\frac{I_{new}}{I_{old}} = \left(\frac{A_{new}}{A_{old}}\right)^2$ $\frac{0.5I_{old}}{I_{old}} = \left(\frac{A_{new}}{4y}\right)^2$ $A_{new} = 2.8y$
18	**B**	For image to be resolved, $\theta > \frac{\lambda}{\quad} = \frac{500\times10^{-9}}{\quad} = 2.0\times10^{-5}$ rad